

Show this to the shopkeeper

"Hello. I am building a small camera-and-laser turntable with two stepper motors. I need these parts. Please check that each one is the exact type written."

ITEM	QTY	PRICE (ESTIMATE)	SHOP	MUST BE
☐ Arduino UNO R3 + USB-B cable	1	₹450–700	Amar shop	CH340 clone is fine. Amazon: "Arduino UNO R3 CH340 with cable".
☐ CNC Shield V3 for UNO	1	₹250–350	Amar shop	It must say "V3" and fit on an UNO. Amazon: "CNC shield V3 Arduino UNO".
☐ 12 mm laser module, red 650 nm, 5 mW, 5 V	1	₹100–200	Amar shop	Only 5 mW. Never a "burning" laser. Amazon: "650nm 5mW laser module 12mm".
☐ Dupont jumper wires, female-female, 20 cm	1	₹60–100	Amar shop	One pack. For the laser.
☐ Fuse holder 5×20 mm + 3 A fuses + rocker switch	1	₹60–120	Amar shop	Safety for the 12 V. Optional but cheap.
☐ Heatsinks for A4988 (if the drivers have none)	1	₹20–40	Amar shop	Small stick-on type.

Check at the counter

- **CNC shield:** the text on the board says "V3", and it fits on the UNO (it has 4 driver sockets).
- **Laser:** 12 mm wide metal tube, red, 5 mW, 5 V. Ask for the datasheet or label.
- **UNO:** has a USB-B cable. A CH340 clone is fine.

We already have: 2 × A4988 drivers, a 12 V adapter, 2 × NEMA17 motors, a USB camera. Do not buy these again.

Amar shop total: about ₹940–1,510

Screws, nuts and small things

"I need M3 screws and nuts in a kit, some tiny self-tapping screws, and strong glue."

ITEM	QTY	PRICE (ESTIMATE)	SHOP	MUST BE
☐ M3 screw kit (6 to 30 mm) with nuts and washers	1	₹250–400	Hardware shop	Covers every screw below. Amazon: "M3 screw nut assortment kit".
☐ M2 × 6 self-tapping screws	1	₹20–40	Hardware shop	Four are used for the camera board.
☐ Super glue + rubber feet (or non-slip tape)	1	₹60–120	Hardware shop	Three feet under the tripod legs.

What the screw kit must contain (counts we use)

SCREW	HOW MANY	USED FOR
M3 × 8 mm	11	4 pan motor · 4 tilt motor · 1 hub set screw · 1 head set screw · 1 laser clamp
M3 × 10 mm	3	tilt mount onto the hub
M3 × 12 mm	8	4 cap to tower · 4 tower to base
M3 nut	6	4 under the base · 1 hub · 1 head
M2 × 6 mm self-tapping	4	camera board

A kit of 6 to 30 mm M3 screws has all of these with spares. Buy a few extra M3 × 8.

Hardware shop total: about ₹330–560

Amazon (only if a shop does not have it). Search words: "Arduino UNO R3 CH340", "CNC shield V3 Arduino UNO", "650nm 5mW laser module 12mm", "M3 screw nut assortment kit", "Dupont jumper wire female female". Check the delivery date: we need the parts before printing ends.

Not needed any more: No bearings, belts or pulleys. No angle sensors, no splitter. No breadboard, no soldering. No metal shaft hubs (the hubs are printed and clamp with a set screw).

Print job sheet

"Please print these parts in PETG. The files are attached. Print the test coupon first so we can check the fits."

PART	QTY	SIZE MM	WEIGHT	SUPPORTS	HOW TO LAY IT
Tripod base 01_tripod_base.stl	1	190 × 167 × 6	52 g	No	Flat on the bed, as shown. Three legs and the square pad. Four nut pockets on the underside.
Tower sleeve 02_tower_sleeve.stl	1	74 × 74 × 46	66 g	No	Foot flange on the bed, open end up. Holds the pan motor. A cable slot near the bottom.
Tower cap 03_tower_cap.stl	1	60 × 60 × 4	16 g	No	Flat on the bed. The pan motor screws to its underside.
Pan hub 04_pan_hub.stl	1	34 × 34 × 16	9 g	No	Small end down, wide end up. Clamps on the pan shaft. Slot for one M3 nut.
Tilt mount 05_tilt_mount.stl	1	52 × 47 × 51	26 g	No	Plate down, wall and ribs standing up. The tilt motor screws to its wall.
Head 06_head.stl	1	58 × 40 × 70	29 g	Yes, only under the camera plate	Stand it on the rear of the clamp, laser tube pointing up. Camera on the front, laser tube on top, clamps on the tilt shaft.
Test coupon (print FIRST) 00_test_coupon.stl	1	100 × 46 × 8	28 g	No	Flat on the bed. Checks every fit in about an hour. See page "Do the parts fit?".

Settings: Material PETG (PLA is fine for the first test). Layer 0.2 mm. 4 walls. Infill 25%. No brim needed.

Files: one STL per part in the pack (Plan_Urena_Tower_print_pack.zip). STEP files are inside too, in case the shop needs them.

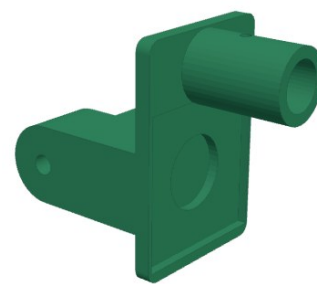
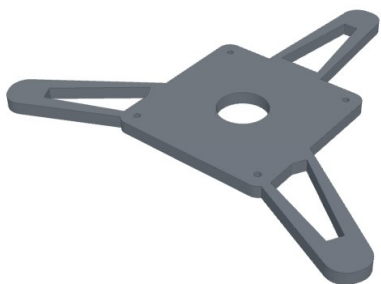
Total: 198 g for the six parts (+ 28 g for the coupon). About 14–20 hours on a normal printer (estimate).
Biggest part: the tripod base, 190 × 167 mm, fits a 220 mm bed.

At a print shop

About ₹990–1,980 (₹5–10 per gram, ask for a quote) + ₹140–281 for the coupon.

On a friend's printer

About ₹240–400 of PETG for the six parts (estimate).



Left: tripod base. Right: head (camera plate on the front, laser tube on top).